

Fig. 3: 1+1 linear protection switching

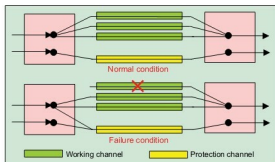


Fig. 4: 1:N linear protection switching

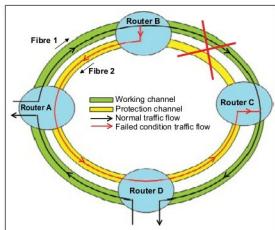


Fig. 5: Unidirectional ring protection

100 per cent redundancy, as there is a spare line for each working line. This architecture is simple for implementation and results in fast restoration. But, its major drawback is the wastage of bandwidth, since no useful traffic flows through the redundant paths.

1:N APS architecture. Economic considerations have led to the preferential use of 1:N architecture, particularly for long-distance paths. In this case, a single back-up line protects several working lines (Fig. 4). When the primary path/channel fails, the two ends of the affected path are

switched over to the back-up line/channel. During normal operation, no traffic or low-priority traffic is sent through the protection/redundant path.

When any failure occurs (such as, fibre-cut), both the source and destination switch onto the redundant or alternate path. Here, all switching is revertive, which means, the traffic shifts to the working line as soon as the failure is corrected. The reserve circuits can be used for lower-priority traffic, which is simply interrupted if the circuit is needed to replace a failed working line. Although network utilisation is better in this architecture, it requires signalling overhead and also results in slower restoration.

Ring protection

A ring is the simplest and most cost-effective way of linking a number of network elements. The greater the communication bandwidth carried by transmission media, the greater the cost advantages of ring structures as compared to linear structures. Various protection mechanisms are available for ring architecture, such as, unidirectional, bi-

directional, revertive and non-revertive connections.

Unidirectional mode means that the two network elements (NEs) choose independently which circuit to receive, without negotiation. In all the modes, the working and protection interfaces receive the same payload from add-drop multiplexer (ADM), but only one is selected or currently active. The deselected interface is held in a 'line protocol is down' state and is completely removed. Only the selected interface actually processes the payload.

In bidirectional mode, receive and transmit channels are switched as a pair. But transmit and receive channels are switched independently in the unidirectional mode. For example, in bidirectional mode, if receive channel on the working interface has a failure event, both transmit and receive channels are switched.

In revertive connection, the hardware switches back to the working line automatically after repair of the working line or after the elapse of a configured period. In the non-revertive connection, if a failure condition occurs, the hardware switches to the protection line and does not automatically revert to the working line.

Unidirectional rings. In ring topology, traffic is transmitted simultaneously over the working and the protection lines. If there is an interruption, the receiver switches to the protection line and immediately takes up the connection. This switching process is referred to as line switching. A simpler method is to use the so-called path-switching ring in which a backup path is used from the source to its destination to bypass the failure.

Fig. 5 shows the basic principle of APS for unidirectional rings. Let us assume that there is an interruption in the circuit between the network elements, say, router B and C. In this situation, node adjacent to the fault will detect the condition and start the APS protocol. K1 and K2 bytes of the SDH frame indicate the current status of the APS connection and convey bridge requests, node information, type of failure, etc to the affected nodes. Each node detecting a fault sends an APS request to the node to which it was connected in the direction of fault. The connection is therefore switched to the alternative path in network elements.

Bidirectional rings. In this network architecture, connections between network elements are bidirectional. Often, bidirectional line-switched ring (BLSR) is used in which the overall capacity of the network can be split up for several paths, each with one bidirectional working line. For unidirectional rings,